

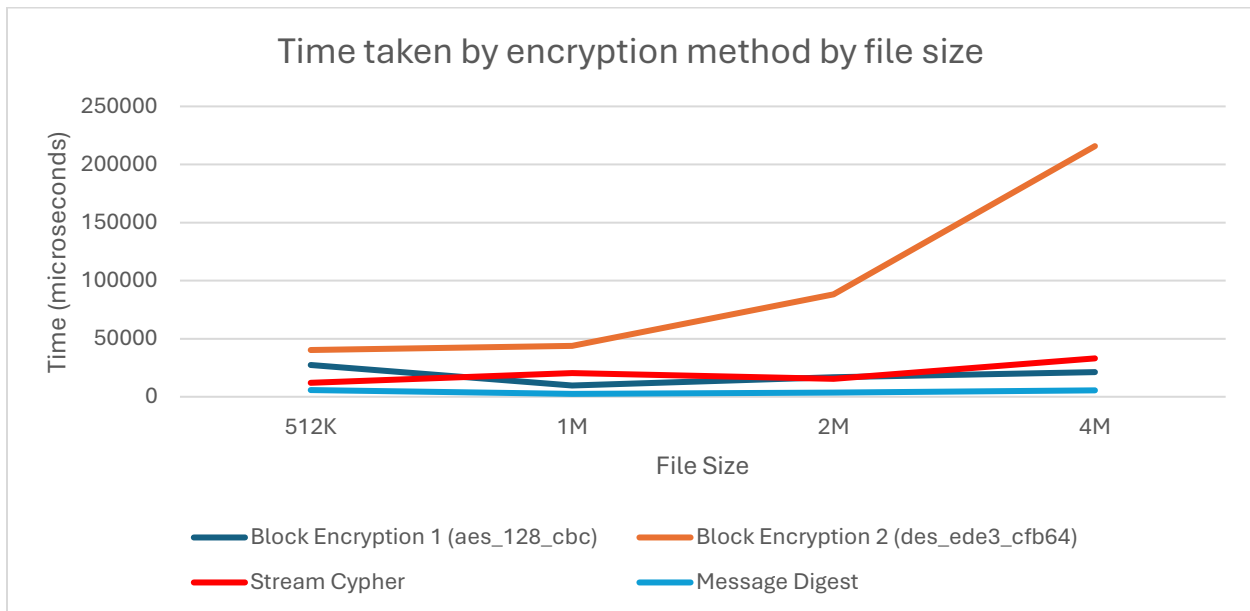
Assignment 4 Report

By: Group 2

Data Table:

File Size	Block Encryption 1 (aes_128_cbc)	Block Encryption 2 (des_edes3_cfb64)	Stream Cypher	Message Digest
512K	27369	40269	12020	5788
1M	9713	43733	20477	2500
2M	16854	88153	15563	3537
4M	21354	215785	33053	5513

Graph:



Report:

Compare the 2 symmetric encryptions. Explain your observations. –

The two symmetric encryption algorithms tested were AES-128-CBC and 3DES-CFB. AES performed significantly faster than 3DES, especially as the file size increased. This is expected because 3DES performs the DES encryption process three times for every block of data, which increases computational cost. AES is also a newer algorithm designed to be efficient and is optimized on modern processors. Because of this, AES scales better as file size increases.

Compare block vs. stream ciphers. Explain your observations. –

The stream cipher and the block cipher (AES) showed similar performance, with the stream cipher sometimes performing slightly faster. This is because stream ciphers encrypt data one byte at a time, while block ciphers must process fixed-size blocks and perform additional operations such as padding and cipher modes.

In general, stream ciphers can be faster due to simpler operations, while block ciphers often provide stronger and more widely trusted security mechanisms.

Compare symmetric encryption and digest generation. Explain your observations. –

The digest generation was faster than the symmetric encryption methods. This is expected because hashing only performs a one-way transformation of the data, while encryption must perform multiple reversible transformations using keys and cipher modes.

Encryption requires more processing steps such as key scheduling and multiple transformation rounds, while hashing focuses only on producing a fixed-size output. Because of this, digest generation is typically faster than encryption.